STAT 516: Discrete Random Variables

Lecture 5: CDF/pmf, Medians, Basic moments

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Basic properties of expectations

Example): $X = \begin{cases} c, & \text{prob } 1 \\ x \neq c, & \text{prob } 0 \end{cases}$ $\mathbb{E}(X) = c \cdot 1 = c$

1 ▶ For a finite constant c s.t. $P(X = c) = 1$ we have $\mathbb{E}\, X = c$

2 ▶ For X and Y on the same sample space Ω with finite expectations, if $P(X \leq Y) = 1$, we have $\mathbb{E}\, X \leq \mathbb{E}\, Y$ $\mathbb{E}X(X) = \sum x_i p_X(x_i) \leq \sum y_i$. etc

3 ▶ If X has a finite expectation, and $P(X \geq c) = 1$, $\mathbb{E}\, X \geq c$; if $P(X \leq c) = 1$, $\mathbb{E}\, X \leq c$ Follows from 2. Just select $Y = c$ with prob 1.

Linearity of expectations; expectation of a function

- Let $X_1, \ldots, X_n$ defined on the same Ω and $c_1, \ldots, c_k$ are any real valued constants. Then

$$\mathbb{E}\left(\sum_{i=1}^{k} c_i X_i\right) = \sum_{i=1}^{k} c_i \mathbb{E}\left(X_i\right)$$

specifically:

$\mathbb{E}(X_1+X_2) = \mathbb{E}(X_1)+\mathbb{E}(X_2)$

$\mathbb{E}(cX) = c\mathbb{E}(X)$

- If X is defined on Ω and $Y = g(X)$, and if $\mathbb{E}\, Y$ exists,

$$\mathbb{E}\left(Y\right) = \sum_{\omega} Y(\omega)P(\omega) = \sum_{x} g(x)p(x)$$

Expectation of independent random variables

- Let $X_1, \ldots, X_k$ be independent random variables; if each expectation exists, we have

$$\mathbb{E}\left(X_1 X_2 \ldots X_k\right) = \mathbb{E}\left(X_1\right)\mathbb{E}\left(X_2\right)\cdots\mathbb{E}\left(X_k\right)$$

Example

$x = 2, 3 \ldots 12$

- Let X be the sum of two rolls when a fair die is rolled twice
- Check that pmf of X is $p(2) = p(12) = \frac{1}{36}$;
  $p(3) = p(11) = \frac{2}{36}$ etc.
- $\mathbb{E}\, X = 2\frac{1}{36} + 3\frac{2}{36} + \cdots = 7$ $+ 11\left(\frac{2}{36}\right) \cdot 12\left(\frac{1}{36}\right)$
- Alternatively, define X_1 the face on the first roll, X_2 the face on the second roll, then Possible x: $\{1, 2, 3, 4, 5, 6\}$

$$\mathbb{E}\, X = \mathbb{E}\,(X_1 + X_2) = \mathbb{E}\, X_1 + \mathbb{E}\, X_2 = 3.5 + 3.5 = 7$$

First roll $E(X_1) = 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) \ldots + 6\left(\frac{1}{6}\right)$

$E(X_1) = (1+2+3+4+5+6)\left(\frac{1}{6}\right) = \frac{21}{6} = 3.5$

Second roll $E(X_2) = E(X_1) + E(X_2) = 2 \cdot (3.5) = 7$

Example

- Let a fair die be rolled 10 times and X be the sum of these rolls
- The pmf is hard to write down exactly; but if X_i is the face on the ith roll,

$$\mathbb{E}\,X = \mathbb{E}\,(X_1 + X_2 + \cdots + X_{10})$$
$$= \mathbb{E}\,(X_1) + \mathbb{E}\,(X_2) + \cdots + \mathbb{E}\,(X_{10}) = 3.5 \times 10 = 35$$

Linearity of expectation does not require independence!

Use of indicator variables to compute expectations of discrete random variables

A RV $I_A \begin{cases} 1, & w \in A \\ 0, & w \notin A \end{cases}$

success

$\uparrow$

$P(I_A) = 1 \cdot P(A) = P(A)$

- Let $c_1, \ldots, c_m$ be constants and $A_1, \ldots, A_m$ some events
- Let X be an integer valued random variable $X = \sum_{i=1}^m c_i I_{A_i}$; then

$$\mathbb{E}\, X = \sum_{i=1}^{m} c_i P(A_i)$$

- Coin tosses - let a fair coin be tossed n times with the probability of success p
- The number of successes is $X = \sum_{i=1}^n I_{A_i}$ for obvious A_i
- $\mathbb{E}\, X = \sum_{i=1}^n P(A_i) = np$

Example

- The matching problem: let the number at location i be $p(i)$; define X as the number of locations such that $p(i) = i$
- Again, define obvious A_i and $X = \sum_{i=1}^{n} I_{A_i}$ $A_i = \{$ There is a match at the i-th location $\}$
- For any i, $P(A_i) = \frac{(n-1)!}{n!} = \frac{1}{n}$
- Thus, $\mathbb{E}\, X = \sum_{i=1}^{n} P(A_i) = 1$

Example

"what is the chance I have not seen the faces in the first n rolls?"

$P(X>n)$: faces hasn't happened yet after n rolls.

faces comes after roll n.

"Every roll from 1 to n was NOT the face.

- Let a fair die be rolled n times; X is the number of faces that never show up in these n rolls
- Let A_i be the event that ith face is missing; $X = \sum_{i=1}^{6} I_{A_i}$
- For any i, $P(A_i) = \left(\frac{5}{6}\right)^n$
- $\mathbb{E} X = \sum_{i=1}^{6} P(A_i) = 6 \times \left(\frac{5}{6}\right)^n$
- If $n = 10$, this is about 0.97

Independent

$\frac{5}{6} \cdot \frac{5}{6} \cdot \frac{5}{6} \cdots$

$= [P(X=1) + P(X=2) + P(X=3) + \ldots] +$
$[P(X=2) + P(X=3) + P(X=4) + \ldots] +$
$[P(X=3) + P(X=4) + P(X=5) \ldots + \ldots]$

$= 1 \cdot P(X=1) + 2P(X=2) + 3P(X=3) \ldots$

$= \sum_{k=1}^{\infty} k \cdot P(X=k) = \mathbb{E}(X)$

Tail sum method

The Expected # of Steps is the sum of the prob. that you're still rolling at each step.

Use Tail Sum Method to calculate $\mathbb{E}(X)$ of discrete non-negative random variable by adding up probabilities that X is greater than each integer.

- Let X take values $0, 1, 2, \ldots$. Then,

Prob of continued failure

$$\mathbb{E}X = \sum_{n=0}^{\infty} P(X > n)$$

prob that you're still waiting after n rolls

- Let p be the probability of success in a Bernoulli trial; how long do we wait on average for the first success?
- If X is the number of trials needed than $X > n$ means that the first n trials all resulted in tails
- $\mathbb{E}X = \sum_{n=0}^{\infty} P(X > n) = \sum_{n=0}^{\infty}(1-p)^n = \frac{1}{p}$

Ex): If $p = 1/5$

If $\mathbb{E}(X) = 5$ trials needed

Example

- Suppose a couple will have children until they have one child of each sex. How many children can they expect to have?
- Let X be the childbirth at which they have a child of each sex for the first time
- If the births are independent, the probability that a childbirth is a boy is p, we have

Girl

Boy (Complement of Boy)

$$P(X > n) = p^n + (1-p)^n$$

- Therefore,

$\sum_{n=2}^{\infty} p^n$ $\sum_{n=2}^{\infty} (1-p)^n$

$$\mathbb{E}\,X = 2 + \sum_{n=2}^{\infty}[p^n + (1-p)^n] = 2 + \frac{p^2}{1-p} + \frac{(1-p)^2}{p} = \frac{1}{p(1-p)} - 1$$

$$\frac{(1-p)^2}{\cancel{1-}(1-p)}$$

Variance of discrete random variables

- **Variance** of X is
$$\sigma^2 = \mathbb{E}\left[(X-\mu)^2\right]$$
- The standard deviation is $\sigma = +\sqrt{\sigma^2}$
- Possible alternative is $\mathbb{E}\left|X-\mu\right|$...can show that $\sigma \geq \left|X-\mu\right|$

- $Var\left(cX\right) = c^2 var\left(X\right)$
- $Var\left(X + k\right) = Var\left(X\right)$ for any real k
- $Var\left(X\right) \geq 0$; $Var\left(X\right) = 0$ iff $X = \mu$ w.p.1
- $Var\left(X\right) = \mathbb{E}\left(X^2\right) - \mu^2$

$= \mathbb{E}(X^2 - u^2)$

QED $= [X^2 - 2uX + u^2]$

$= \mathbb{E}(X^2) - 2u\underset{u}{\mathbb{E}(X)} + u^2$

Moments of a discrete random variable X

- For a positive integer $k \geq 1$ we call $\mathbb{E}(X^k)$ a **kth moment** of X
- $\mathbb{E}(X^{-k})$ is a **kth inverse moment** of X
- If $\mathbb{E}[|X|^3] < \infty$, the **skewness** of X is

$$\beta = \frac{\mathbb{E}[(X-\mu)^3]}{\sigma^3}$$

- The skewness measures how symmetric the distribution if X is; e.g. for $X \sim N(0,1)$, $\beta = 0$
- If $\mathbb{E}[X^4] < \infty$, the **kurtosis** of X is

$$\gamma = \frac{\mathbb{E}[(X-\mu)^4]}{\sigma^4} - 3$$

- The kurtosis is always ≥ -2 and is equal to zero for a normal distribution; it measures how "spiky" the distribution is around its mean

Example

- Let X be the sum of two independent rolls of a fair die. We know that $\mathbb{E}(X) = 7$
- $\mathbb{E}(X^2) = \frac{329}{6}$ and $Var(X) = \mathbb{E}(X^2) - \mu^2 = \frac{329}{6} - 49 = \frac{35}{6} = 5.83$

Example: Variance in the Matching Problem

- Let X be the number of locations where match occurs when n numbers are rearranged in a random order
- We know that $\mathbb{E}(X) = 1$ for any n; moreover, recall representation $X = \sum_{i=1}^{n} I_{A_i}$
- Now,

$$\mathbb{E}(X^2) = \sum_{i=1}^{n} P(A_i) + 2 \sum_{1 \le i < j \le n} P(A_i \cap A_j)$$

$$= n \times \frac{1}{n} + 2 \binom{n}{2} \frac{(n-2)!}{n!} = 1 + 1 = 2$$

- Thus, $Var(X) = 2 - 1 = 1$ for any n
- Think of which distribution might approximate well the number of matches...

Variance of a sum of independent random variables

- Let $X_1, \ldots, X_n$ be independent random variables;

$$Var\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} Var\,(X_i)$$

- As a corollary (and a very important one!), if $\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n}$, and $\sigma^2 = Var\,X_i < \infty$,

$$Var\,(\bar{X}) = \frac{\sigma^2}{n}$$

Chebyshev's and Markov's inequalities

- (Chebyshev's inequality) Let $\mathbb{E}\,X = \mu$ and $Var\,X = \sigma^2$ be finite. For any positive number k we have

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

- (Markov's inequality) Suppose X takes only nonnegative values and $\mathbb{E}\,X = \mu$ is finite. If c is any positive number,

$$P(X \geq c) \leq \frac{\mu}{c}$$

- Chebyshev's inequality is a direct consequence of Markov's inequality

Example

- Chebyshev's and Markov's inequalities are rather conservative
- Let X be the sum of two rolls of a fair die. Recall that $\mu = 7$ and $\sigma = 2.415$
- Choose $k = 2$ in the Chebyshev's inequality; direct calculation gives
$$P(|X - 7| \geq 4.830) = \frac{1}{18} = 0.056$$
- Chebyshev's inequality gives the lower bound of $\frac{1}{4} = 0.25$

Weak law of large numbers

- WLLN is a direct consequence of Chebyshev's inequality
- Let $X_1, \ldots, X_n$ be iid RV's with $\mathbb{E}\, X_i = \mu$ and $Var\, X_i = \sigma^2 < \infty$.
- For any $\varepsilon > 0$,
$$P(|\bar{X} - \mu| > \varepsilon) \to 0$$
as $n \to \infty$
- A stronger version is the strong law of large numbers (SLLN) that says that
$$P(\lim_{n\to\infty} \bar{X} = \mu) = 1$$
- The only condition needed is that $\mathbb{E}\,|X_i|$ be finite

Truncated Distributions

- Examples: planet observations; reported car accidents
- Let X be a discrete random variable with pmf $p(x)$; let A be a fixed subset of its values
- **The distribution of** X **truncated to** A has the pmf

$$p_A(y) = \frac{p(y)}{P(X \in A)}$$

for any $y \in A$ and 0 if $y \notin A$

- The mean of a truncated distribution is

$$\mu_A = \frac{\sum_{y \in A} y p(y)}{\sum_{y \in A} p(y)}$$

Example

- Let $P(X = n) = \frac{1}{2^n}$ for $n = 1, 2, \ldots$; we only observe X if $X \leq 5$
- The truncation set is $A = \{1, 2, 3, 4, 5\}$ and

$$p_A(y) = \frac{(1/2)^y}{\sum_{y=1}^{5}(1/2)^y} = \frac{2^{5-y}}{31}$$

  for $y = 1, 2, \ldots, 5$
- Check that its mean is 1.71 which is less than $\sum_{n=1}^{\infty} n \times \frac{1}{2^n} = 2$
- (Chow-Studden inequality). For any RV X and finite real constants a and b, let $U = \min(X, a)$ and $V = \max(X, b)$. Then,

$$Var\,(U) \leq Var\,(X);\, Var\,(V) \leq Var\,(X)$$